Homework 5

ECON 470, Spring 2026

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1. Mean Uncompensated Care by Year

Table 1: Mean hospital uncompensated care (mm) by year, 2011–2018

Year	Uncomp. Care	Hospitals
2011	30.740000	4044
2012	33.130000	4124
2013	33.670000	4096
2014	29.480000	4103
2015	29.220000	4080
2016	39.610000	4043
2017	38.000000	4116
2018	40.660000	4049

Uncompensated care rose from 30.74mm per hospital in 2011 to 33.67mm in 2013. It then fell to 29.48mm in 2014 and further to 29.22mm in 2015, consistent with the ACA's Medicaid expansion taking effect. The mean rose sharply to 39.61mm in 2016 and continued climbing through 2018, driven in part by hospitals with extreme values pulling up the mean.

2. Uncompensated Care Over Time by Expansion Status

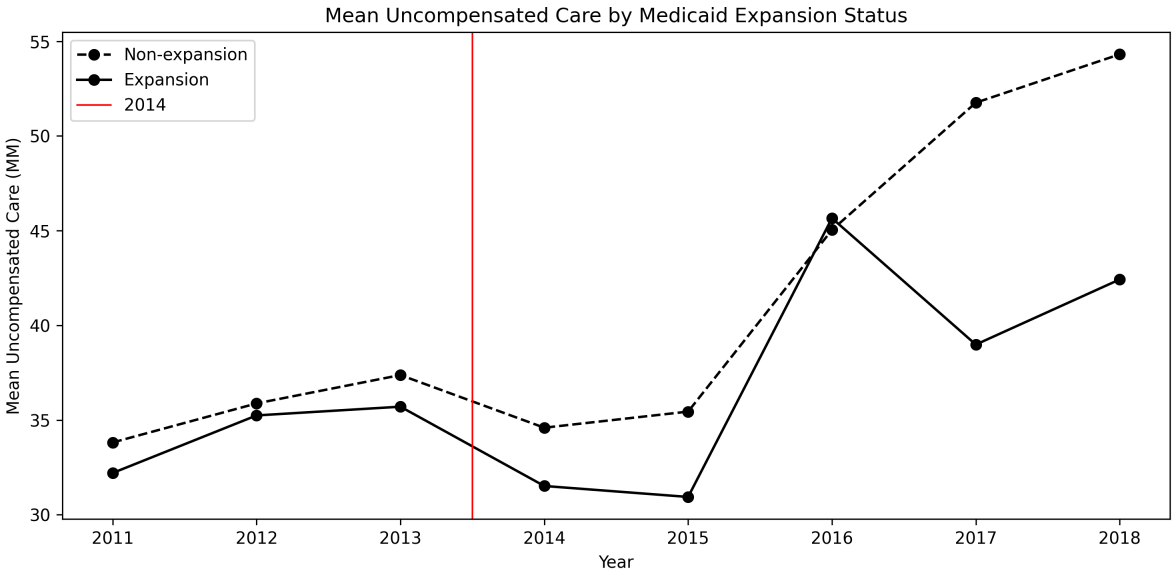

Figure 1: Mean uncompensated care over time for 2014 expanders vs never-expanders

Figure 1 presents mean uncompensated care by expansion status. Both groups show similar pre-2014 trends, with a clear divergence after 2014 as expansion states see larger reductions in uncompensated care. In 2013, expanders averaged 35.71mm and never-expanders averaged 37.38mm. By 2015, expanders fell to 30.94mm while never-expanders stood at 35.45mm.

3. 2x2 Difference-in-Differences Table

Table 2: 2x2 DD table of mean uncompensated care (mm), pre (2012) vs post (2015)

year	Pre (2012)	Post (2015)
Non-expansion	35.880000	35.450000
Expansion	35.240000	30.940000

Table 2 presents the 2x2 DD table. The DD estimate is -3.87 million dollars, suggesting that Medicaid expansion reduced hospital uncompensated care.

4. Policy Discussion

Several forces were at work during this period. The most direct is Medicaid expansion itself, which covered adults up to 138 percent of the federal poverty level. Hospitals could bill for care they previously wrote off entirely. But the ACA's insurance exchanges also pulled in uninsured people in non-expansion states, which partly explains why never-expanders also saw declines. The broader economic recovery from the 2008 recession reduced the share of patients unable to pay, and S-10 reporting standards shifted around 2012-2013, so some of the measured change reflects accounting practices rather than actual care delivered. We expect expansion to reduce uncompensated care because insured patients generate reimbursable claims. Hospitals get paid instead of absorbing the cost.

5. Standard DD Regression

The standard DD estimate of the effect of Medicaid expansion on uncompensated care is -5.022mm (SE 4.962) on 23344 observations. The negative sign is consistent with Medicaid expansion reducing uncompensated care, though the estimate does not reach statistical significance. Full results are presented alongside the TWFE estimates in Table 4.

6. TWFE with Hospital and Year Fixed Effects

Table 3: DD Estimates with TWFE

	Standard DD	TWFE
(Intercept)	35.698	
	(3.351)	
Post 2014	8.511	
	(4.245)	
Expand	-1.302	
	(3.921)	
Post x Expand	-5.022	-4.918
	(4.962)	(8.231)
Num.Obs.	23,344	23,265
R2	0.000	0.37
R2 Within		0.0
Std.Errors		by: state

Table 4 presents the TWFE results alongside the standard DD. The TWFE estimate with hospital and year fixed effects accounts for time-invariant hospital characteristics and common year shocks. The standard DD gives a Post x Expand coefficient of -5.022mm (SE 4.962) while the TWFE estimate is -4.918mm (SE 8.231) on 23265 observations. Results are qualitatively similar, though the point estimate differs due to the additional controls.

7. TWFE with All States

Table 4: DD Estimates with Staggered Treatment

	Standard DD	TWFE	Staggered
(Intercept)	35.698		
	(3.351)		
Post 2014	8.511		
	(4.245)		
Expand	-1.302		
	(3.921)		
Post x Expand	-5.022	-4.918	-2.161
	(4.962)	(8.231)	(5.558)
Num.Obs.	23,344	23,265	32,554
R2	0.000	0.37	0.387
R2 Within		0.0	0.0
Std.Errors		by: state	by: state

Table 5 presents results including all states with staggered treatment timing. Including all states shifts the Post x Expand estimate from -4.918mm to -2.161mm (SE 5.558) on 32554 observations. This can reflect the standard problem with TWFE under staggered adoption, where already-treated states serve as implicit controls for later-treated ones and can contaminate the estimate if treatment effects differ across cohorts.

8. Event Study: 2014 Expanders vs Never-Expanders

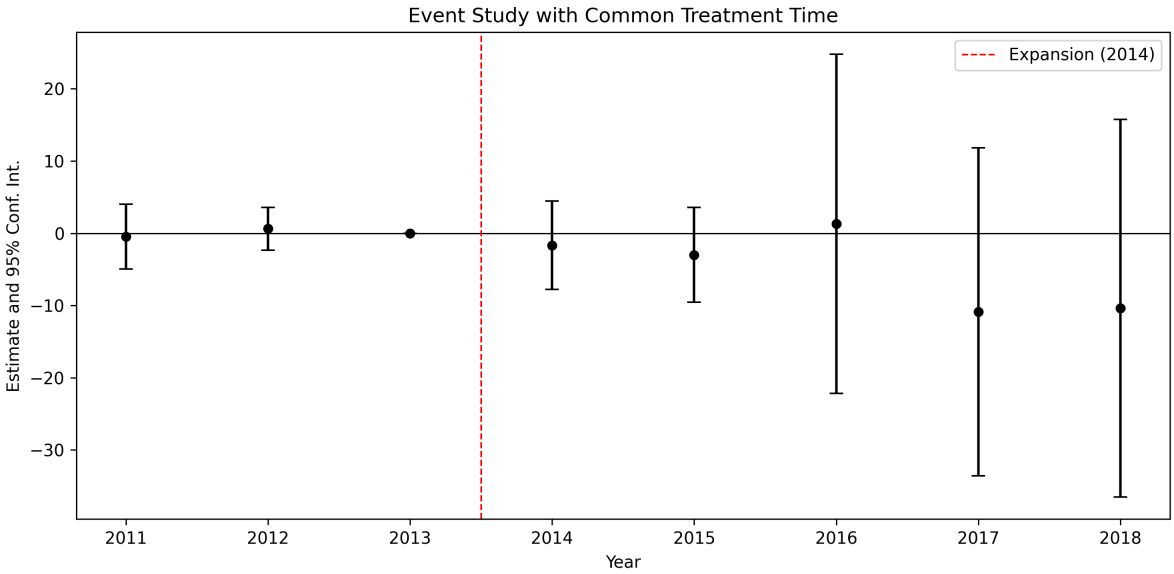

Figure 2: Event study: effect of Medicaid expansion on uncompensated care, 2014 expanders vs never-expanders (ref: 2013)

Figure 2 presents event study estimates for 2014 expansion. Pre-2014 coefficients near zero support the parallel trends assumption. Post-expansion coefficients turn negative and grow in magnitude, consistent with a persistent reduction in uncompensated care among expanding states relative to never-expanders.

9. Event Study: All States (Event Time)

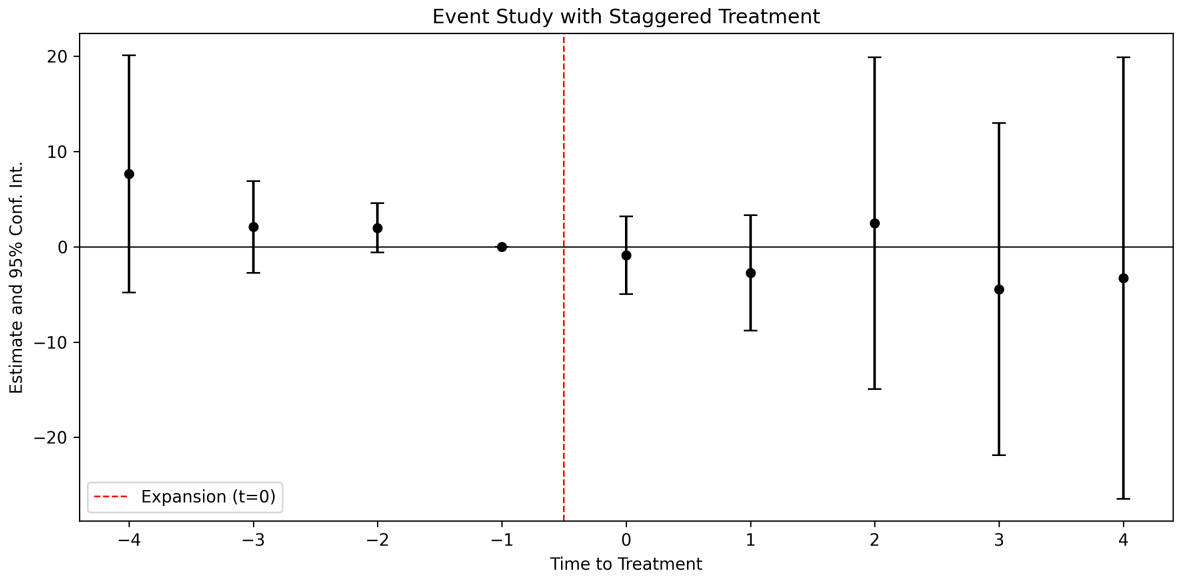

Figure 3: Event study: effect of Medicaid expansion on uncompensated care, all states on event time (ref: t=-1)

Figure 3 presents the event study with staggered treatment. Time 0 denotes the year of expansion, with t-1 as the reference period. Endpoints are binned at t-4. Pre-expansion coefficients are near zero, supporting parallel trends. Post-expansion estimates turn negative and grow, consistent with a persistent reduction in uncompensated care.

10. Summary and TWFE Limitations

Across Questions 5 through 9, results consistently point toward Medicaid expansion reducing uncompensated care, though none reach statistical significance. The simple DD gave -5.022mm (SE 4.962), the TWFE on the 2014 sample gave -4.918mm (SE 8.231), and adding all states produced -2.161mm (SE 5.558). They are all negative and in the expected direction.

The event study in Q8 raises some concern about parallel trends. Pre-period coefficients are not uniformly close to zero, suggesting expanding and non-expanding states may have been on somewhat different trajectories before 2014. Post-expansion, coefficients turn negative and remain so, consistent with a persistent reduction in uncompensated care.

The core limitation of TWFE with staggered adoption is that it uses already-treated states as controls for later-treated ones. If early and late expanders experienced different effect sizes, this contaminates the estimate and can produce negative weights on some underlying 2x2 DDs. Other estimators that we briefly talked about in class address this by estimating cohort-specific effects before aggregating, which avoids the problem of using treated units as controls.